Robot Vision: Mock Exam 5 (Solutions)

Chapters 2 to 12 · 90 minutes · no documents · 40 points

Part A: Multiple choice

22 points, about 35 min

A1 A network must predict three real values with no restriction on their range. Which output design is appropriate? (ch. 2)

(a) ✓ **Three units with identity activation and a squared-error loss.**
(b) A softmax over three units with cross-entropy.
(c) Three sigmoid units with binary cross-entropy.
(d) One sigmoid unit with squared error.

Regression of unbounded values: identity output.

A2 What is the main purpose of momentum in SGD? (ch. 2)

(a) To switch off units randomly.
(b) ✓ **To smooth oscillating updates and accelerate movement in consistent directions.**
(c) To compute the Hessian of the loss.
(d) To reduce the number of parameters.

Add μ times the previous step.

A3 A ReLU layer has fan-in M. What standard deviation does He initialization use for its weights? (ch. 3)

(a) $1/M$
(b) $2/M$
(c) ✓ $\sqrt{2/M}$
(d) $\sqrt{M/2}$

$2/M$ is the variance; the std is its square root.

A4 Why is the learning rate often decreased during training? (ch. 3)

(a) To increase the number of parameters.
(b) Because a large rate is needed at the end.
(c) ✓ **To make fast progress early and finer adjustments later.**
(d) To make the loss noisier.

Large steps first, fine steps later.

A5 What is the receptive field of a unit? (ch. 4)

(a) All the output maps of the layer.
(b) The region removed by pooling.
(c) The set of weights of its filter.
(d) ✓ **The region of the input that can affect the output of that unit.**

It grows with depth.

A6 Which statement about translation equivariance and invariance is correct? (ch. 4)

(a) Standard convolution guarantees invariance to any shift.
(b) ✓ **Equivariance: shifting the input shifts the output; invariance: the output does not change.**
(c) Equivariance: the output does not change; invariance: it shifts.
(d) Both mean that the output does not change.

Convolution guarantees equivariance; pooling gives approximate invariance.

A7 How does head-orientation estimation with a CNN differ from ordinary classification? (ch. 5)

(a) It predicts one class among three.
(b) It uses a softmax over angles.
(c) ✓ **It predicts continuous values (roll, pitch, yaw) instead of class labels.**

(d) It needs no training data.

Regression: linear output, squared error.

A8 What is the main advantage of dilated convolution for segmentation? (ch. 5)

(a) It reduces the number of channels.
(b) ✓ **A larger receptive field without extra weights and without downsampling.**
(c) It removes the need for nonlinearities.
(d) It doubles the number of weights.

Responses at all positions.

A9 In TCAV, what represents a concept, and what is the TCAV score? (ch. 6)

(a) A superpixel; the score is its Shapley value.
(b) A single neuron; the score is its maximum activation.
(c) ✓ **A direction in a layer's activation space; the score is the fraction of class samples with a positive directional derivative along it.**
(d) A filter; the score is its norm.

Sensitivity, not causality.

A10 A saliency map barely changes after the model weights are randomized. What does this show? (ch. 6)

(a) The model is very robust.
(b) ✓ **The explanation does not really describe what the trained model has learned.**
(c) The model has no parameters.
(d) The explanation is especially faithful.

Model-parameter randomization sanity check.

A11 Why can LP-FT be better than direct fine-tuning? (ch. 7)

(a) ✓ **Training the new head first, with the backbone frozen, prevents a random head from distorting the pretrained features.**
(b) It trains the backbone first and the head never.
(c) It removes the pretrained features.
(d) It uses more labeled data.

Also better out-of-distribution.

A12 What is the difference between unsupervised domain adaptation and domain generalization? (ch. 7)

(a) ✓ **Domain adaptation uses unlabeled target-domain data during training; domain generalization does not use target data.**
(b) They are the same.
(c) Domain generalization uses labeled target data.
(d) Domain adaptation uses no source data.

UDA: labeled source + unlabeled target.

A13 Which statement about out-of-distribution (OOD) inputs is correct? (ch. 8)

(a) ✓ **A classifier can give a high softmax confidence to an OOD input.**
(b) Softmax confidence is a reliable OOD detector.
(c) A classifier always gives a low confidence to OOD inputs.
(d) OOD inputs cannot be classified at all.

It must pick one of its known classes.

A14 What does quantization-aware training (QAT) do? (ch. 8)

(a) It quantizes a trained model without any retraining.
(b) ✓ **It keeps float weights but simulates quantization in the forward pass during training.**
(c) It trains a small student from a teacher.
(d) It removes whole filters.

The rounding gradient uses the straight-through estimator.

A15 What does the recurrent part of RAFT do? (ch. 9)

(a) It segments the objects.
(b) It predicts the next frame.
(c) It processes the video frame by frame over time.
(d) ✓ **It iteratively refines the optical-flow estimate for the same pair of images.**

RAFT's iterations are not time steps.

A16 Nodes are renumbered, with A and X permuted consistently. What should happen to the outputs? (ch. 9)

(a) ✓ **Node-level outputs follow the permutation; graph-level outputs stay the same.**
(b) Both stay the same.
(c) Both change unpredictably.
(d) Graph-level outputs follow the permutation; node outputs stay the same.

$A' = PAP^\top$, $X' = PX$.

A17 Why is positional information added to the tokens of a Transformer? (ch. 10)

(a) ✓ **Self-attention without it is permutation equivariant and does not know the order of the tokens.**
(b) It makes the model recurrent.
(c) It replaces the value vectors.
(d) It speeds up the softmax.

Sinusoidal or learned.

A18 How do Swin and MaxViT make attention cheaper? (ch. 10)

(a) Swin uses grid attention; MaxViT uses shifted windows.
(b) Both remove attention.
(c) ✓ **Swin uses local windows shifted between layers; MaxViT combines local block attention with sparse grid attention.**
(d) Both compute full global attention at every layer.

Cost of full attention grows as N^2.

A19 How is a large language model pre-trained? (ch. 11)

(a) By classifying manually labeled sentences.
(b) By reinforcement learning only.
(c) ✓ **By predicting the next token on large amounts of text, which is self-supervised.**
(d) By predicting images from text.

Generation is autoregressive.

A20 How does CLIP classify an image "zero-shot"? (ch. 11)

(a) It generates a caption and searches it.
(b) It uses a softmax with one unit per ImageNet class.
(c) ✓ **It compares the image embedding with the embeddings of texts like "a photo of a {class}" and picks the most similar.**
(d) It fine-tunes on a few labeled images.

No classifier trained on the target labels.

A21 What does pix2pix require, and what does its discriminator see? (ch. 12)

(a) No images; the discriminator sees noise.
(b) Unpaired images; the discriminator sees only the output.
(c) Paired images; the discriminator sees only the input.
(d) ✓ **Paired images; the discriminator sees the input together with the output.**

Adversarial + L1 loss.

A22 What is the main advantage of a latent diffusion model? (ch. 12)

(a) It works only on pixels.
(b) It needs no denoising network.
(c) ✓ **Diffusion runs in the smaller latent space of an autoencoder, which lowers the cost.**
(d) It removes the need for any training data.

Conditions enter by cross-attention.

Part B: Short answers

12 points, 3 per question, about 35 min

B1 What does Adam combine, and why is it popular? (ch. 3)

Momentum (moving average of the gradients) and RMSProp (moving average of the squared gradients, one step size per weight), with bias correction. It works well with little tuning.

B2 Why is the output of an FCN too coarse for segmentation, and how does U-Net fix it? (ch. 5)

The backbone downsamples (output stride 32). U-Net concatenates high-resolution encoder features with the upsampled decoder features to recover fine details.

B3 How does PointNet handle a set of points? Explain invariance and equivariance here. (ch. 9)

A shared MLP processes each point (equivariant), max pooling combines them (symmetric, so the global feature is invariant). Classification is invariant to the point order; segmentation is equivariant.

B4 What are the three stages of training a modern LLM? (ch. 11)

Pre-training by next-token prediction on large text (self-supervised); instruction SFT on instructions with desired answers; preference optimization (RLHF, DPO).

Part C: Calculations

6 points, 2 per question, about 15 min

C1 How many parameters does a fully connected layer with 100 inputs and 10 outputs have? (ch. 2)

$100 \cdot 10 + 10 = \mathbf{1010}$.

C2 How many weights (no bias) does a 1×1 convolution from 256 to 64 channels have? (ch. 5)

$1 \cdot 1 \cdot 256 \cdot 64 = \mathbf{16\,384}$.

C3 A dilated 3×3 convolution has dilation rate 2. What is the span it covers? (ch. 5)

$k + (k-1)(r-1) = 3 + 2 \times 1 = \mathbf{5}$, i.e. 5×5, still with 9 weights.